

HTML Tags — Class Note

1. Tags With and Without Closing Tags

A. Tags that DO NOT need a closing tag

These are called Void Elements.

Tag	Purpose
 	Line break
<hr>	Horizontal/thematic break
	Display an image
<input>	Create an input field
<meta>	Page metadata
<link>	Link an external resource, such as CSS
<area>	Clickable area inside an image map
<base>	Set a base URL
<col>	Define a table column
<embed>	Embed external content
<source>	Provide media source
<track>	Provide subtitles/captions for media
<wbr>	Possible word-break position

Examples:

```
<br>

<input type="text">
<hr>
<meta charset="UTF-8">
<link rel="stylesheet" href="style.css">
```

Note: You may also see:

```
<input type="text" />

```

The / before > is optional in HTML. <input> and <input /> are normally treated the same in HTML.

B. Tags that use a closing tag

For beginner-level HTML, remember this simple rule:

If a tag is NOT in the void-element list above, write its closing tag.

Examples:

```
<html> ... </html>
<head> ... </head>
<title> ... </title>
<body> ... </body>

<h1> ... </h1>
<h2> ... </h2>
<h3> ... </h3>
<h4> ... </h4>
<h5> ... </h5>
<h6> ... </h6>

<p> ... </p>
<pre> ... </pre>
<div> ... </div>
<span> ... </span>

<b> ... </b>
<strong> ... </strong>
<i> ... </i>
<em> ... </em>
<u> ... </u>

<a> ... </a>

<ul> ... </ul>
<ol> ... </ol>
<li> ... </li>

<table> ... </table>
<caption> ... </caption>
<thead> ... </thead>
<tbody> ... </tbody>
<tfoot> ... </tfoot>
<tr> ... </tr>
<th> ... </th>
<td> ... </td>

<form> ... </form>
<label> ... </label>
<textarea> ... </textarea>
<select> ... </select>
<option> ... </option>
<button> ... </button>
<fieldset> ... </fieldset>
<legend> ... </legend>

<style> ... </style>
<script> ... </script>

<header> ... </header>
<nav> ... </nav>
<main> ... </main>
<section> ... </section>
<article> ... </article>
<aside> ... </aside>
<footer> ... </footer>

<iframe> ... </iframe>
<audio> ... </audio>
<video> ... </video>
```

Easy Rule:

Normal Element → Opening Tag + Content + Closing Tag

```
<p>Hello</p>
```

Void Element → Only one tag

```
<br>
<img>
<input>
```

2. Important HTML Tags for This Web Technologies Course

Students do not need to memorize every HTML tag. The following tags are the important ones for this course.

Basic HTML Page

Tag	Purpose
<!DOCTYPE html>	Tells browser that the document uses HTML5
<html>	Root of the HTML document
<head>	Contains page information/settings
<title>	Browser tab title
<meta>	Metadata such as character encoding
<body>	Contains visible webpage content

Note: <!DOCTYPE html> is a declaration, not an HTML element/tag.

Text and Content

Tag	Purpose
<h1> to <h6>	Headings
<p>	Paragraph
 	Line break
<hr>	Horizontal/thematic break
<pre>	Preserves spaces and new lines
	Bold text
	Important text
<i>	Italic/alternate text
	Emphasized text
<u>	Underlined/annotated text
<div>	Block container/group

	Inline container/group
--------	------------------------

Links and Images

Tag	Purpose
<a>	Creates a hyperlink
	Displays an image

Example:

```
<a href="https://www.aiub.edu/">
  
</a>
```

Lists

Tag	Purpose
	Unordered/bullet list
	Ordered/numbered list
	List item

Example:

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Tables

Tag	Purpose
<table>	Creates a table
<tr>	Table row
<th>	Table heading cell
<td>	Table data cell
<caption>	Table title/caption
<thead>	Groups table header rows
<tbody>	Groups main table rows
<tfoot>	Groups table footer rows

Basic structure:

```
<table>
  <tr>
    <th>Name</th>
    <th>ID</th>
  </tr>

  <tr>
    <td>Jim</td>
    <td>123</td>
  </tr>
</table>
```

Forms — Very Important for This Course

Tag	Purpose
<form>	Creates a form
<label>	Gives a label to an input
<input>	Creates an input field
<textarea>	Multi-line text input
<select>	Creates a dropdown list
<option>	Creates an option inside a dropdown
<button>	Creates a button
<fieldset>	Groups related form fields
<legend>	Gives a title to a fieldset

Example:

```
<form>
  <label for="userId">User ID:</label>
  <input type="text" id="userId" name="userId">

  <button type="submit">Submit</button>
</form>
```

CSS and JavaScript Related Tags

Tag	Purpose
<link>	Connects an external CSS file
<style>	Writes CSS inside an HTML page
<script>	Adds JavaScript to the page

Example:

```
<link rel="stylesheet" href="style.css">

<script src="script.js"></script>
```

Page Structure Tags

Tag	Purpose
<header>	Top/header section
<nav>	Navigation section
<main>	Main page content
<section>	Section of a page
<article>	Independent article/content
<aside>	Side content
<footer>	Bottom/footer section

Important: <input> Types

These are NOT different HTML tags.

For example:

```
<input type="text">
<input type="password">
<input type="email">
<input type="number">
<input type="radio">
<input type="checkbox">
<input type="file">
<input type="submit">
```

The HTML tag is always:

```
<input>
```

Only the value of the type attribute changes.

Quick Memory

Page Structure:
html → head + body

Text:
h1-h6, p, pre, br

Link/Image:
a, img

List:
ul/ol → li

Table:
table → tr → th/td

Form:
form → label + input/textarea/select/button

CSS:
link / style

JavaScript:
script